Regime-Aware Temporal Granularity for Causal Video World Models: When Does Adaptivity Pay?

Md Jisan Hossen[a]

[a]Department of Artificial Intelligence, Nanjing University of Information Science and Technology (NUIST), Nanjing, China

ARTICLE INFO

Keywords:
causal video generation
video world models
autoregressive diffusion
adaptive compute
temporal granularity
regime map

ABSTRACT

Causal video generators produce a rollout one temporal chunk at a time, so spending expensive per-frame compute only on hard segments looks attractive. We show that this idea rests on three *preconditions* — cost, signal and endpoint — and that all three are *regime properties* rather than universal facts. On a matched two-GPU study (Tesla T4 and A100-80GB) of a causal diffusion backbone we (i) measure the per-forward cost against the schedule break-even and find the same mode pair on opposite sides of the inequality; (ii) show that hand-crafted context statistics cannot predict coarse-unsafe segments (AUROC 0.491), that an *intrinsic* probe on the generator's own prefill activations reaches 0.798 on one GPU and *inverts* to 0.143 on the other, and that only a *multi-feature* model over the probe and a sampler convergence residual transfers across both (0.842 and 0.833); and (iii) show that an endpoint independent of the compared modes is mandatory, because the internal reference reverses between GPUs. We then derive a *regime map* that states exactly when adaptivity pays — only when the cheap mode is cheaper *and* worse — for a stated objective. Across two GPUs and three independent endpoints — latent (both GPUs), decoded-pixel and temporal flicker (T4) — our operating points fall outside that region: the coarse mode was never worse in quality, so under the stated objective the optimal policy is a *fixed* mode and a router is unnecessary. We also report a training-free seam repair that reduces the switch-boundary discontinuity by 25% while improving fidelity, quantify drift propagation, and record three boundaries: no single signal transfers across GPUs, drift is linear and policy-insensitive, and a pixel-only decision-model critic is at chance. All results, figures and audited result files are public.

1. Introduction

Interactive video world models — game engines, robot simulators, embodied agents and augmented reality — must synthesise frames under a latency budget on whatever accelerator is available. A causal (autoregressive) video generator emits the rollout one temporal chunk at a time, and chunks are not equally hard. It is therefore natural to spend *fine* per-frame computation on difficult segments and a cheaper *coarse* multi-frame mode elsewhere, deciding before each segment is generated.

This idea is appealing but conditional. Our earlier study [5] framed it as resting on three *preconditions*: (P1) a cheap mode must be genuinely cheaper on the deployment hardware at the chosen schedule; (P2) a signal available *before* generation must predict which segments are unsafe for the cheap mode; and (P3) the quality claim must be certified by an endpoint that is independent of the modes being compared. That study measured the preconditions on a single GPU and reported a conditional trade-off.

*Corresponding author.
jisanahammed1952@gmail.com (M.J. Hossen)
ORCID(s): 0009-0004-9571-1450 (M.J. Hossen)

This paper asks the question that follows: *when does adaptivity actually pay?* We answer with a matched two-GPU study (Tesla T4 and A100-80GB) of a causal diffusion backbone, and we make four contributions.

- **A two-GPU regime map** (§4) that partitions the (cost-ratio, quality-gap) plane into three policies — fixed fine, fixed coarse and adaptive — and states precisely when a router is worth deploying. The adaptive region is defined by a conjunction: the cheap mode must be *cheaper and worse*.
- **An intrinsic signal — and the limit of single signals.** A probe on the generator's own prefill activations predicts coarse-unsafe segments at AUROC 0.798 with no external oracle, against 0.491 for hand-crafted context statistics. The same probe *inverts* on the second GPU (0.143, CI $[0.018, 0.368]$), while a free sampler *convergence residual* (the ratio of the first to the last denoising-step update magnitude) merely *falls to chance* there (0.760 on T4, 0.404 on A100, whose interval $[0.223, 0.603]$ contains 0.5). Only a *multi-feature* model over the two transfers across both GPUs (0.842 and 0.833).
- **A training-free seam repair.** Switching modes creates a measurable discontinuity at the boundary. Blending the first latent frame of the new segment toward the boundary by 25% reduces the seam by 25% *and* improves fidelity, with no training.
- **A boundary analysis.** We show, and record, that signals are regime properties: the intrinsic probe *inverts* on the second GPU ($0.798 \rightarrow 0.143$). Drift is linear and not predictable from the policy, and a decision-model critic shown only pixels is at chance. Each boundary is stated with the experiment that produced it.

A summary of the programme is shown in Fig. 1. Throughout, we report the *measured* limit of each naive formulation rather than treating it as a failure; the boundaries are the paper's design output.

Relationship to the prior study. The prior preprint [5] measured the three preconditions on a *single* GPU with an *external* decision-model oracle. This paper is a distinct study: it extends the measurement to two GPUs, replaces the external oracle with an intrinsic signal, adds the convergence residual and the seam repair, and derives the regime map. The audit summarised in §3 (the *report*) is an unpublished artifact of the accompanying repository; no text or figure of the prior preprint is reused here.

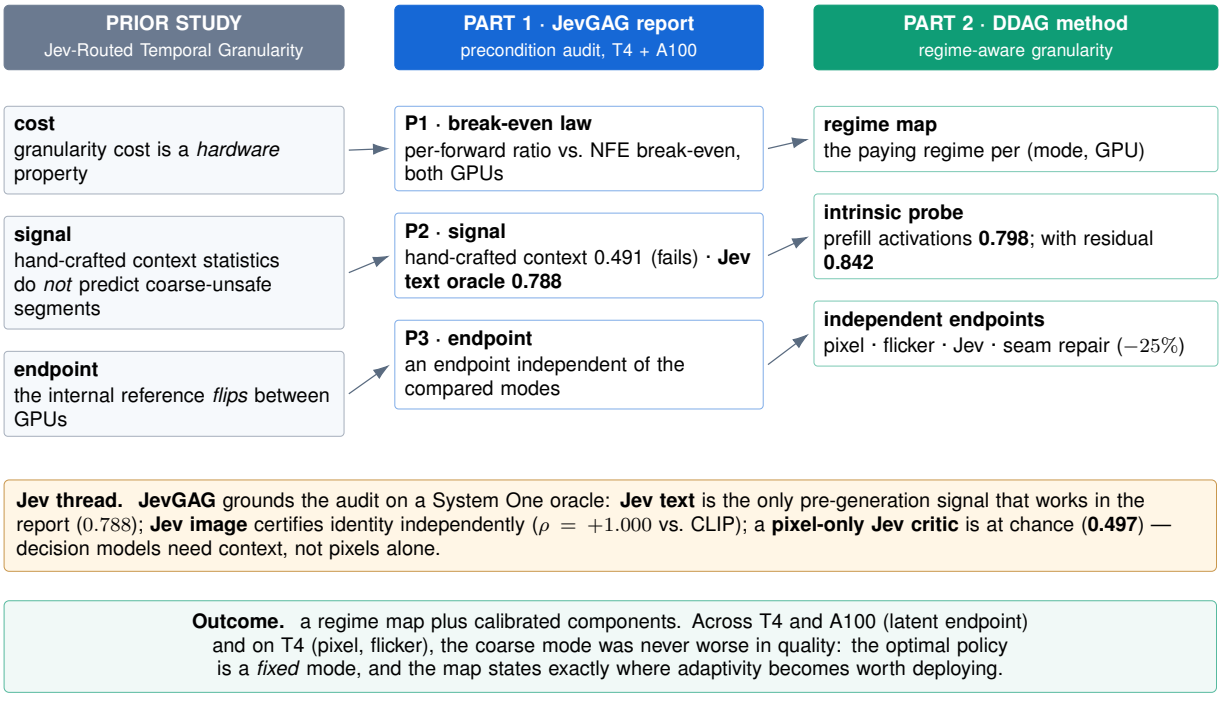


Figure 1: The programme. The earlier study established three preconditions; the report in this work measures them on two GPUs; the method turns them into a regime map and calibrated components. Each row is one precondition — the boundary found and its resolution. The decision-model (Jev) thread that grounds the audit is highlighted. The regime-aware method of Part 2 is *DDAG* (the repository code name for the regime-aware method).

2. Related Work

Our contribution sits at the intersection of five threads. We review each in turn and state, for each, how this study differs.

Causal and autoregressive video generation. Streaming video generators produce a rollout chunk by chunk, and the train–test gap between teacher forcing and self-rolling is a central difficulty [6, 31, 37]. Recent recipes distil few-step causal diffusion models for interactive use [30, 33], and autoregressive transformer generators scale similar ideas [17]. Our backbone is one of this family; we treat it as a black box and study its *granularity*, not its training.

Adaptive compute and routing. Adaptive computation was introduced for recurrent networks [4], early exiting for classification [20], and dynamic routing [21]. In generative video, recent work allocates computation by step or by token: dynamic step allocation [8], learned flexible chunking [14], pruning and routing [23], layer reuse [1], and caching of attention or flow state [15, 18, 22, 26]. Adaptive tokenisation changes the *representation* [10, 24, 28, 34]. Our work differs in object and in method: we route the *temporal granularity* of a fixed backbone, and — crucially — we do not assume that routing pays; we characterise the regime in which it does.

Internal probes and verification. Ensembles of internal classifiers and early-exit policies read a network's own intermediate representations [19, 25, 27, 29, 35]. We adapt this idea to *generation-mode safety*: a probe on a diffusion transformer's prefill activations predicts whether a cheap mode is unsafe for the upcoming segment. Verification-first policies are related to speculative decoding [2, 9, 13], but speculation there operates over tokens; here it operates over the temporal-chunking axis.

Evaluation of generative video. Video generative models are evaluated along many axes, including temporal flickering [7, 32], artefact taxonomies [36], and compression-aware consistency [38]. We adopt a multi-endpoint protocol: a latent reference, a decoded-pixel reference and a temporal-flicker statistic, so that a claim cannot be an artefact of one metric.

Decision models. System One decision models return typed, calibrated answers rather than generated text [3, 11, 12]. We refer to the decision model used here and in the earlier study as *Jev* (a System One model, `jev-1.14.0`). The earlier study used Jev as an off-device routing oracle; here we test it as a *critic* and report the boundary we find, consistent with independent evidence that pixel-only quality prediction with such models is unreliable [16].

3. Background: the three preconditions

Setting. A causal generator with two granularity modes is rolled out over T segments. A *fine* mode emits one latent frame per chunk; a *coarse* mode emits K frames jointly. A policy chooses a mode per segment. We write c_f, c_c for the per-segment cost of the two modes and q_f, q_c for a quality functional (lower is better). We write s_a/s_b for a *schedule pair*: the fine generator runs a denoising steps per latent frame and the coarse generator b steps per chunk. The one-frame-per-chunk fine mode is written $c_{1\text{-}1}$ and the three-frame-per-chunk coarse mode $c_{3\text{-}3}$; both run two steps by default, so the matched two-step pair is s_2/s_2, and a two-step-fine / one-step-coarse pair is s_2/s_1.

The three preconditions.

P1 (cost). The cheap mode must satisfy $c_c < c_f$ on the deployment hardware at the chosen schedule. Because a coarse forward is heavier than a fine forward, the condition is a comparison of the measured per-forward ratio against the *break-even* set by the number of function evaluations of each schedule, $\rho_{\text{be}} = \text{NFE}_f/\text{NFE}_c$. Coarse is cheaper iff the measured per-forward ratio π is *below* ρ_{be}.[1]

P2 (signal). A quantity computable *before* a segment is generated must separate segments for which the cheap mode is unsafe. The earlier study found that hand-crafted context statistics fail this test and that an off-device decision model passes it.

[1] For the matched two-step pair ($\text{NFE}_f = 34$, $\text{NFE}_c = 12.67$) $\rho_{\text{be}} = 2.68$; for the two-step-fine / one-step-coarse pair ($\text{NFE}_c = 8.33$) $\rho_{\text{be}} = 4.08$. The prior study's single-step break-even (2.33) is a different schedule and is not used for any claim in this paper.

P3 (endpoint). The quality claim must be certified by a measure that is independent of the two modes. The earlier study found that an internal reference reverses direction between GPUs.

Matched design and label. For every prompt and segment we generate *both* modes from the same boundary and the same noise, so that the only difference is the generator. A segment is *coarse-unsafe* when the coarse segment is farther from a high-quality reference (a four-step fine rollout) than the fine segment, and the label is $y = \mathbb{1}[\, d(m_c) > d(m_f)\,]$, with $d(\cdot)$ the endpoint distance of a generated segment $m \in \{\text{fine}, \text{coarse}\}$. The symbol q denotes the *expectation* of d, so $q_f = \mathbb{E}[d(m_f)]$.

The report. We call the audit of the preconditions the *JevGAG report*, and the regime-aware method of §4 *DDAG* (the repository code name for the regime-aware method); the title frames the same method as *regime-aware*. On T4 the report finds that coarse is *not* cheaper for a two-step coarse schedule, that hand-crafted context statistics reach only 0.491 AUROC while the Jev text oracle reaches 0.788, and that the internal reference flips between GPUs. On A100 the cost ordering reverses. The report also shows that a two-threshold controller with a hysteresis band and a forced-recovery window stabilises switching, and that switching leaves a measurable *seam*. Full details, code and audited result files are in the repository accompanying this paper.

4. Method: regime-aware granularity

4.1. The regime map

Let $r = c_c/c_f$ be the cost ratio and $\Delta = q_c - q_f$ the quality gap, with $\Delta > 0$ meaning the cheap mode is worse.

Objective. A policy is evaluated under the constrained objective

$$\text{minimise } \mathbb{E}[\text{DiT seconds}] \quad \text{subject to} \quad \mathbb{E}[d] \le \mathbb{E}[d \mid \text{fixed fine}], \tag{1}$$

i.e. spend as little as possible without degrading the endpoint relative to always generating fine. Under Eq. (1) the plane is partitioned into three policies:

$$\text{policy} = \begin{cases} \text{fixed fine}, & r > 1 \quad \text{(the cheap mode is dearer)}, \\ \text{adaptive}, & r < 1 \ \wedge \ \Delta > 0 \quad \text{(cheaper and worse)}, \\ \text{fixed coarse}, & r < 1 \ \wedge \ \Delta \le 0 \quad \text{(cheaper and no worse)}. \end{cases} \tag{2}$$

Equation (2) is the method's core: a router is worth deploying *only* in the middle region. Outside it, one mode dominates and the routing machinery is pure overhead. Figure 2 plots our operating points in this plane. At two of our three

operating points the cost-optimal and quality-optimal fixed modes coincide; at T4 s_2/s_2 they do not, because the fine mode is cheaper while the coarse mode is closer to the reference. In every case the adaptive region is empty — the coarse mode is either dearer ($r > 1$) or no worse ($\Delta \leq 0$) — which is why the regime map is a statement about the regime rather than about a particular objective.

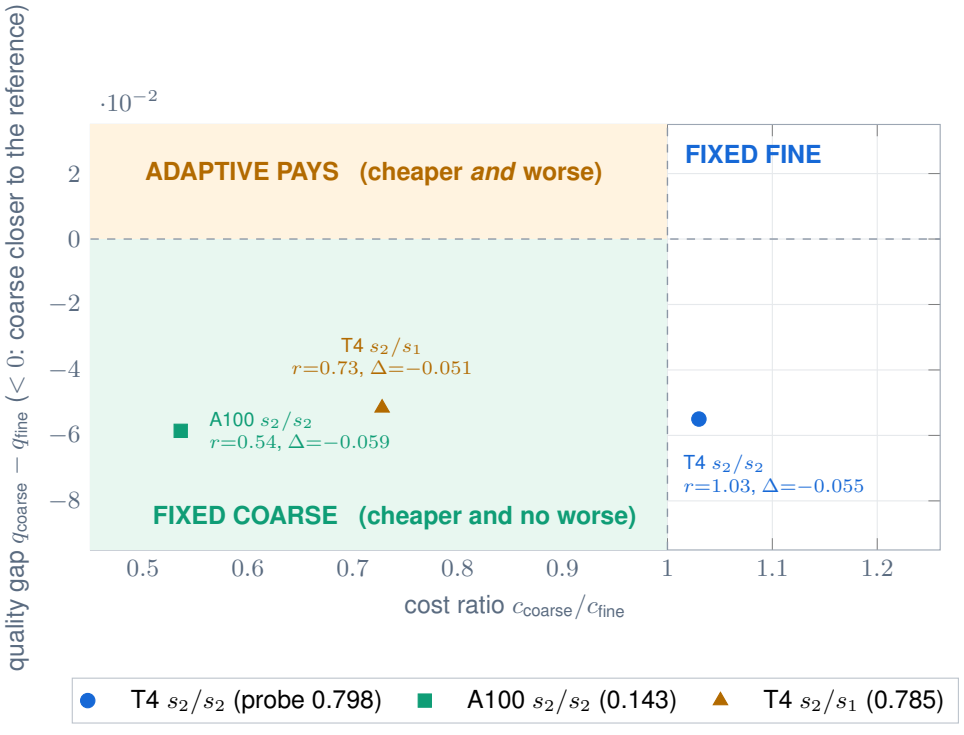


Figure 2: The regime map. The adaptive region (upper left, shaded) is empty at every operating point we measured: across both GPUs the coarse mode was never worse in quality, so the optimal policy is a fixed mode. Markers are labelled with the schedule pair and the intrinsic-probe AUROC at that point.

4.2. An intrinsic pre-generation probe

At the decision point for a segment, the generator has already encoded the boundary context. We capture the pooled output of each transformer block of the single prefill forward on the boundary and fit a logistic probe on a chosen block. The probe is a *pre-generation* quantity: it uses only activations the rollout has already computed, requires no external service, and adds no measurable latency. The probe is evaluated with leave-one-prompt-out cross-validation, so no segment is scored by a model that has seen its own prompt.

4.3. A convergence residual

A distilled few-step sampler has no explicit error estimate, but its update magnitudes are observable. Let $v_1, \ldots, v_S$ be the norm of the predicted velocity at each denoising step of a segment. We define the *convergence residual*

$$\kappa = v_1 / v_S, \tag{3}$$

the ratio of the *first* to the *last* step magnitude: a segment whose initial update is large relative to its final one is a segment the coarse mode has not yet resolved. Because κ is computed from quantities the sampler already produces, it is free. Two cautions apply, and both matter for reproducibility. First, κ must be scored by its *direct* AUROC: passing a single scalar through a logistic regression re-learns its sign and silently reports max(AUROC, 1−AUROC), which hides an inverted signal. Second, a scalar and its reciprocal carry exactly complementary information: $\mathrm{AUROC}(v_S/v_1) = 1 - \mathrm{AUROC}(\kappa)$ holds identically, so the reciprocal v_S/v_1 is *inverted by construction* on any data (direct AUROC 0.240 on T4). The direction of a convergence statistic therefore cannot be read off the quantity itself; the sign must be fixed by definition and checked by direct AUROC.

4.4. Calibrated gating

Given a score u that increases with expected unsafety, a gate escalates a segment to the fine mode when $u > \tau$. Two regimes are distinguished. In *routing* the gate chooses the mode before generation; in *escalation* the cheap mode is always generated and the fine mode is recomputed when the gate fires. Let c_p be the *incremental* cost of obtaining the score once per segment. For the intrinsic probe $c_p \approx 0$: the probe reads the prefill forward the rollout performs anyway (§4.2), so it adds no measurable latency, whereas a score that needs a separate forward would cost $c_p = c_f$. Escalation then costs $c_c + c_p + \rho c_f$ against c_f for fixed fine, so a saving requires $\rho < 1 - r - c_p/c_f$; routing costs $c_p + (1-\rho)c_c + \rho c_f$ and saves whenever $r < 1 - c_p/\big((1-\rho)c_f\big)$. This is why the cost precondition, not the signal, is the binding constraint.

4.5. Training-free seam repair

When the mode changes at a boundary, the first frame of the new segment is generated by a different generator than the frame before it, and the adjacent-frame distance at that boundary exceeds the interior. We repair it without training: the first latent frame of the incoming segment is blended toward the boundary frame (Eq. (4)),

$$\tilde{x}_1 = (1-\alpha)\, x_1 + \alpha\, b, \tag{4}$$

with b the boundary latent and α a scalar. Because the first frame is blended a fraction α toward the boundary, the boundary seam is $(1-\alpha)$ of its unrepaired value *by construction*; the empirical questions are therefore fidelity and flicker, not the seam itself. We sweep α and report the seam–fidelity frontier.

5. Experimental protocol

Backbone and hardware. A causal distilled diffusion video model (1.3B) at 480×832; a four-step fine rollout is the reference. Two accelerators: a **Tesla T4** (Turing, SDPA attention) and an **A100-80GB** (Ampere, FlashAttention-2). The same prompts, seed, horizon and code are used on both.

Prompts and horizon. Sixteen prompts spanning object, scene, motion and human-action strata; a 57-frame horizon with a shared three-frame coarse prefix and four segments of three latent frames each. Drift experiments use an 81-frame horizon with six segments.

Statistics. Leave-one-prompt-out cross-validation grouped by prompt; prompt-grouped bootstrap confidence intervals (2,000–4,000 resamples); permutation nulls for AUROC. Costs are wall-clock DiT seconds measured with device synchronisation around the sampling call.

Endpoints. Three independent endpoints: (i) latent L_1 to the reference; (ii) *decoded-pixel* L_1 to the reference; (iii) a temporal-flicker statistic, the mean absolute consecutive-frame difference. A claim is accepted only if it holds on the endpoints that are independent of the compared modes.

Reproducibility. Backbone: Causal-rCM Wan2.1-T2V-1.3B, upstream commit `ed3cb14`. Seed 17 throughout. The probe is a logistic model on the pooled output of transformer block 11 of the prefill forward; it is evaluated leave-one-prompt-out. Costs are wall-clock DiT seconds around the sampling call, measured with device synchronisation. Table 1 records the sample size and endpoint coverage of every experiment; all result files, kernels and Colab runners are public.

Table 1: Protocol per experiment. "Prompts" is the number of prompts; each prompt contributes four segments at a 57-frame horizon (six at 81 frames). The decoded-pixel endpoint was run on T4 only.

experiment	GPU(s)	prompts	horizon	endpoints
intrinsic probe / convergence	T4, A100	16	57	latent
seam repair	T4	8	57	latent, pixel, flicker
drift propagation	T4, A100	T4: 32, A100: 8	81	latent
online policy	T4, A100	16	57	latent
intermediate band	T4	16	57	latent
closed loop (48 pairs)	T4	16	57	latent
escalation gate	0-GPU	16	57	— (cost model)
decoded-pixel endpoint	T4	8	57	latent, pixel
decision-model critic	T4	16	57	pixel frames
qualitative	T4	3	57	pixel

6. Results

6.1. Cost is hardware-conditional

Two ratios must be kept apart. The *per-forward* ratio π (seconds per single forward, coarse vs. fine) is a hardware property; the *per-segment cost* ratio $r = c_c/c_f$ used by Eq. (2) depends on the hardware and the schedule. In the matched two-step matrix, $\pi = 2.97$ on T4 and 1.74 on A100 against a break-even of 2.68, so the *same* mode pair sits on opposite sides of the inequality on the two GPUs (the prior study's single-step pair gave 3.01 and 1.60 against 2.33). Consistently, $r = 1.030$ for T4 s_2/s_2 (coarse 3% dearer), 0.728 for T4 s_2/s_1 (coarse 27% cheaper) and 0.536 for A100 s_2/s_2 (coarse 46% cheaper). The cost precondition is thus a property of the (schedule, accelerator) pair, as Eq. (2) requires.

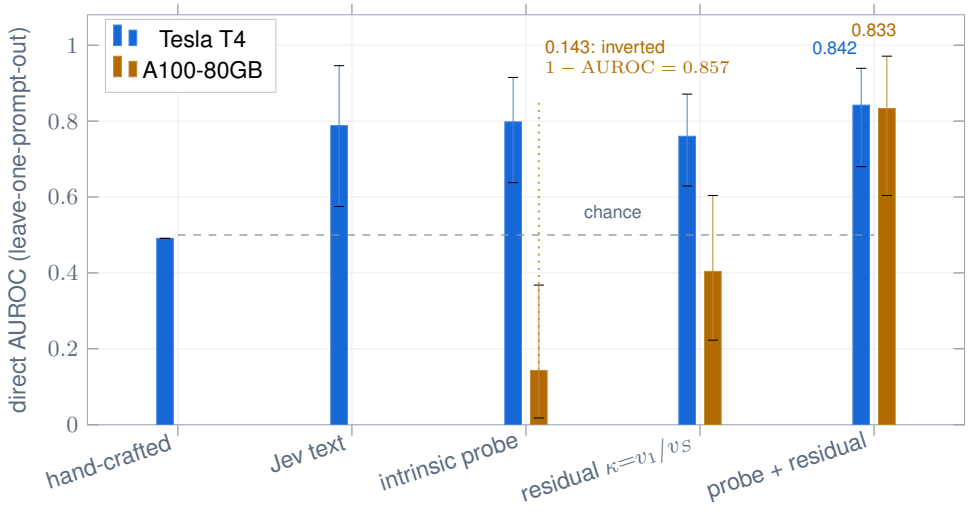


Figure 3: Signal study: direct AUROCs with 95 % prompt-grouped bootstrap intervals ($n = 64$ segments per GPU). Hand-crafted context statistics (0.491) and the Jev text oracle (0.788) are T4-only measurements from the report, and no confidence interval is stored for the hand-crafted statistic, so no whisker is drawn for it. On T4 the intrinsic probe (0.798) and the convergence residual (0.760) both beat hand-crafted, and their combination reaches 0.842. On A100 the probe *inverts* (0.143) and the residual drops to 0.404, yet the combination still reaches 0.833. No single signal transfers; the multi-feature model does.

6.2. An intrinsic signal, and the limit of single signals

Figure 3 summarises the signal study; every value is a *direct* AUROC with a 95 % prompt-grouped bootstrap interval ($n = 64$ segments per GPU). Hand-crafted context statistics reach 0.491; the intrinsic prefill probe reaches **0.798** [0.639, 0.916] on T4, the convergence residual of Eq. (3) reaches **0.760** [0.629, 0.870], and a multi-feature model over the two reaches **0.842** [0.680, 0.938]. The transfer result is the informative part: on A100 the probe *inverts* to 0.143 [0.018, 0.368] and the residual merely *falls to chance*, at 0.404 [0.223, 0.603] (interval contains 0.5), yet the *combined* model still reaches **0.833** [0.605, 0.971] — no single signal transfers, the combination does. Two cautions: the reciprocal v_S/v_1 is inverted *by construction* (AUROC$(v_S/v_1) = 1 -$ AUROC(κ) identically), giving 0.240 [0.130, 0.371] on T4;

and a scalar signal must never be scored through a single-feature logistic, which re-learns the sign and hides any inversion.

6.3. Seam repair

Table 2 reports the repair sweep and Fig. 4 visualises it. A 25 % blend reduces the boundary seam by 25 % (0.614 → 0.460; $\Delta = -0.153$, CI $[-0.162, -0.145]$) and *improves* latent fidelity (0.761 → 0.754; $\Delta = -0.0062$, CI $[-0.0084, -0.0039]$); a hard anchor removes the seam entirely but costs fidelity (0.799). The independent flicker endpoint moves the other way: the coarse mode flickers *less* than fine (0.0148 vs. 0.0189), and the repair raises flicker to 0.0218, above *both*. The repair therefore trades a boundary discontinuity for interior flicker, and we report the trade rather than a single metric. This is why an endpoint independent of the compared modes is mandatory: optimising the seam alone would have hidden the cost.

Table 2: Seam repair (T4, the matched s_2/s_2 pair — fine $c_{1\text{-}1}$ at two steps vs. coarse $c_{3\text{-}3}$ at two steps — on a subset of 8 prompts, 32 matched segments). Lower is better in every column; *fidelity* is the distance to the reference, so a lower value is closer. The 25 % blend reduces the seam and improves fidelity, but the independent flicker endpoint rises — and rises above the fine value — so the repair trades a boundary discontinuity for interior flicker.

variant	boundary seam	fidelity	temporal flicker
none	0.614	0.761	0.0148
blend 0.25	**0.460**	**0.754**	0.0218
blend 0.50	0.307	0.759	–
blend 0.75	0.153	0.774	–
anchor	0.000	0.799	–
fine (no switch)	0.215	0.807	0.0189

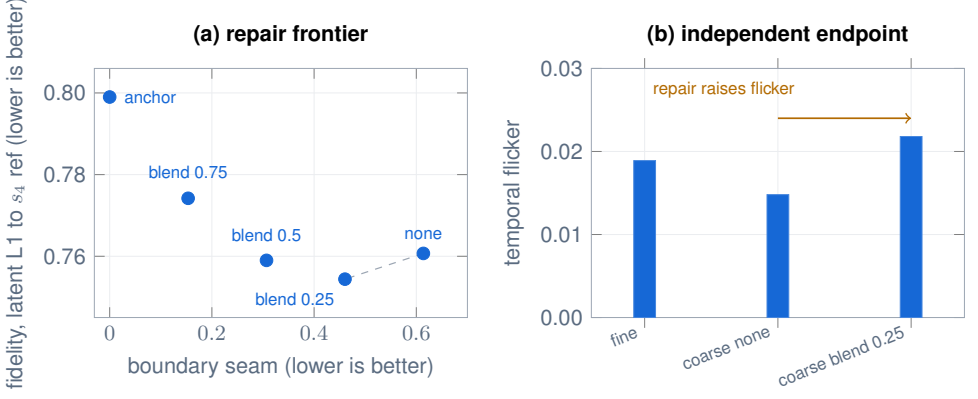


Figure 4: Seam repair. (a) The seam–fidelity frontier: every α lowers the boundary discontinuity, and $\alpha = 0.25$ also improves fidelity, so the repair is a strict improvement on those two axes. (b) The independent flicker endpoint, which moves the other way: the coarse mode flickers *less* than fine, and the repair raises flicker above both. The two panels together are the argument for a multi-endpoint protocol — the metric being optimised is not the metric that certifies.

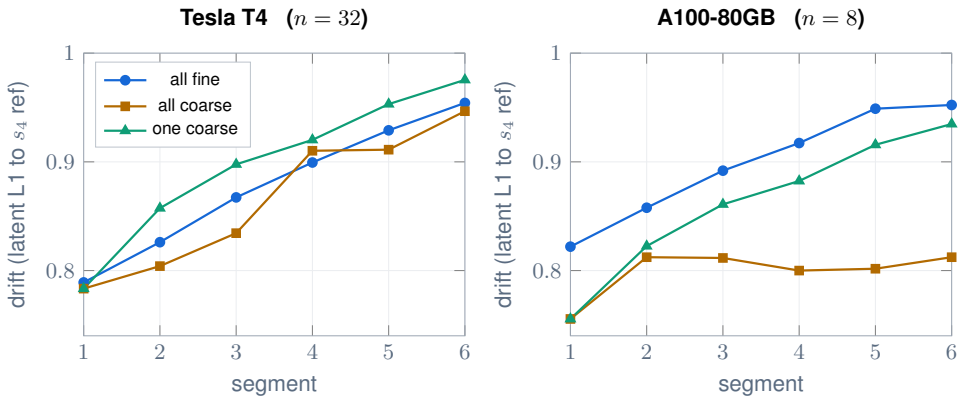


Figure 5: Drift accumulation (81-frame horizon; $n = 32$ prompts on T4, $n = 8$ on A100, six segments each). On T4 a single coarse segment leaves a persistent elevation — significant at four of five tail segments, the fourth only marginally — so the one-coarse chain stays elevated (coarse error and/or switch seam). On A100 the coarse chain lies below the fine chain: the comparison reverses. Drift is linear and not policy-predictable.

6.4. Drift propagates on one GPU and reverses on the other

Figure 5 shows drift trajectories for three chains: all-fine, all-coarse, and one coarse segment followed by fine. On T4 ($n = 32$ prompts) the one-coarse chain remains above the all-fine chain for the rest of the rollout, with a significant elevation at four of five tail segments (+0.031, +0.030, +0.021, +0.024); the fourth excludes zero only marginally (lower bound 0.000), so we treat "four of five" as an upper bound. The one-coarse chain therefore stays elevated (coarse error and/or switch seam); part of the elevation may be a *switch-seam* effect rather than propagated coarse error, because on T4 the all-coarse chain lies *below* the all-fine chain at five of six segments — the coarse mode alone is not uniformly worse. On A100 the coarse chain falls *below* the fine chain; that arm rests on $n = 8$ prompts, which we record as a limitation. Drift is approximately linear, and a transition model fitted on (d_t, mode, t) predicts the next drift level at $R^2 = 0.944$ only because drift is smooth; the granularity choice adds nothing ($R^2 = 0.008$), so a lookahead controller has little headroom.

6.5. The regime map is empty: adaptivity does not pay here

Figure 6 places every verified policy in the cost–quality plane, and Table 3 reports the numbers. On T4 with the two-step pair, fixed fine is cheaper but the coarse mode is closer to the reference (0.752 against 0.807) at 3% extra cost.[2] On A100, and on T4 with a one-step coarse schedule, fixed coarse is both cheaper and closer to the reference. In all three cases the adaptive region of Eq. (2) is empty, so a fixed mode is optimal under Eq. (1): at T4 s_2/s_2 the cost- and quality-optimal fixed modes differ (fine is cheaper, coarse is closer) whereas at the other two operating points they coincide. This is the paper's central finding, and it is a *design rule*: a practitioner computes r and Δ once and deploys accordingly.

[2]The two T4 fine-s_2 baselines (281.2 s in the s_2/s_2 run and 299.9 s in the s_2/s_1 run) come from two separate capture runs of 16 prompts each, and differ by run-to-run wall-clock variance.

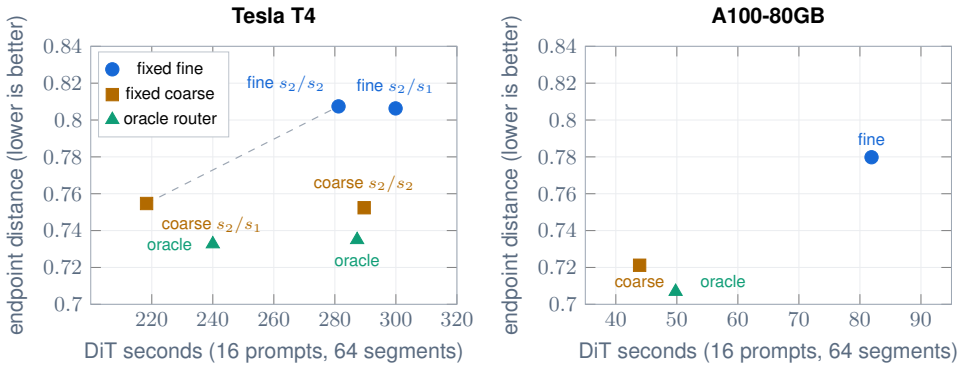


Figure 6: Cost–quality plane, lower-left better, one panel per GPU (16 prompts, 64 segments); both panels share one y-range. At every operating point one fixed mode lies at or beyond the oracle on cost: the oracle router is never *cost*-optimal, although it does buy quality along the frontier. The dashed line in the T4 panel is the fixed-mode frontier between the cheapest coarse point (s_2/s_1) and the fine baseline (s_2/s_2). The adaptive region of Eq. (2) is empty, which is why a fixed policy is optimal under Eq. (1). On T4 s_2/s_2 the cheaper baseline is fine while the closer one is coarse, so cost- and quality-optimality diverge at that point.

An oracle router that knows the true coarse-unsafe label in advance bounds what any signal could buy. It is never *cost*-optimal — it always spends more seconds than the cheapest fixed mode (T4 s_2/s_2: 287.3 vs. 281.2; T4 s_2/s_1: 240.0 vs. 218.3; A100: 49.8 vs. 43.9) — because it must pay both for the decision and for escalated segments. It does buy *quality* (A100: +5.9 s for −0.014 endpoint distance), but it never changes which fixed mode is cost-optimal. The signal decides where on the frontier a deployment sits; the regime decides which mode it should start from.

Table 3: Policies per operating point. "Quality" is the mean endpoint distance (lower is better); "DiT" is total sampling seconds over 16 prompts (64 segments). All values are verified against the stored result files. Bold marks the column minimum within each GPU/schedule group: the best DiT and the best quality, which need not be the same policy. At T4 s_2/s_2 the cheaper mode (fine) is not the closer mode (coarse), so its 0.807 is not the best quality; the oracle router is never cost-optimal because it pays for the decision and for escalated segments.

GPU	policy	DiT (s)	quality	optimal for Eq. (1)
T4, s_2/s_2	fixed fine	**281.2**	0.807	yes
	fixed coarse	289.6	0.752	no (dearer)
	oracle route	287.3	**0.735**	no (adds cost)
T4, s_2/s_1	fixed fine	299.9	0.806	no (dearer)
	fixed coarse	**218.3**	0.755	yes
	oracle route	240.0	**0.733**	no (adds cost)
A100, s_2/s_2	fixed fine	81.9	0.780	no (dearer)
	fixed coarse	**43.9**	0.721	yes
	oracle route	49.8	**0.707**	no (adds cost)

6.6. A third endpoint agrees: decoded pixels

Table 3 rests on the latent endpoint. To check that the conclusion is not an artefact of that metric we re-ran the matched design on T4 for a matched subset of $n = 8$ prompts (32 segments, the same protocol as Table 1) and decoded every segment. This run uses the s_2/s_1 pair of Table 4 (fine $c_{1\text{-}1}$ at two steps, coarse $c_{3\text{-}3}$ at one step); in this smaller run the coarse mode is both cheaper (0.711 of the fine cost — the s_2/s_1 cost ratio *of this* 8*-prompt run*, against $r = 0.728$ in the 16-prompt run of Table 4) and closer by decoded pixels (0.2511 against 0.2686). Three independent endpoints — latent, decoded-pixel and the temporal-flicker statistic — therefore agree that the coarse mode is not worse, which licenses the fixed-mode conclusion. The flicker endpoint is not produced by this run: it is the statistic measured by the seam-repair experiment of §4.5 (T4, the s_2/s_2 pair, 8 prompts, 32 matched segments), so the three endpoints span two schedule pairs and two independent runs. The decoded-pixel endpoint was run on T4 only, which we record as a limitation.

Table 4 collects every measured quantity for the three operating points in one frame.

Table 4: Unified comparison of the three measured operating points in one frame. π is the per-forward ratio and ρ_{be} the schedule break-even (coarse is cheaper iff $\pi < \rho_{\text{be}}$); r is the per-segment cost ratio and $\Delta = q_c - q_f$ the quality gap. "DiT" is the total sampling seconds over 16 prompts. The adaptive region of Eq. (2) is empty at every operating point because the coarse mode is either dearer ($r > 1$) or no worse ($\Delta \leq 0$); at T4 s_2/s_2 the cost-optimal and quality-optimal fixed modes differ (fine is cheaper, coarse is closer).

	T4 s_2/s_2	T4 s_2/s_1	A100 s_2/s_2
per-forward π (ρ_{be})	2.97 (2.68)	3.30 (4.08)	1.74 (2.68)
segment cost ratio r	1.030	0.728	0.536
quality gap Δ	−0.055	−0.051	−0.059
intrinsic probe AUROC	0.798	0.785	0.143[†]
residual κ AUROC	0.760	–	0.404
multi-feature AUROC	0.842	–	0.833
DiT total (s), fine / coarse	281.2 / 289.6	299.9 / 218.3	81.9 / 43.9
optimal policy	fixed fine	fixed coarse	fixed coarse
adaptive region	empty ($r > 1$)	empty ($\Delta \leq 0$)	empty ($\Delta \leq 0$)

[†]inverted: $1 - \text{AUROC} = 0.857$.

6.7. Qualitative behaviour

Figure 7 shows *every* generated image of the study in one frame: all six matched cases (three prompts × two segments), each rendered in the fine, coarse and repaired modes from the same boundary and the same noise, with the amplified difference between fine and coarse. The modes produce visibly *different* content — for the bird prompt the wing pose differs substantially, and the waterfall foam is reorganised — yet no independent endpoint ranks the coarse mode worse. This is the qualitative counterpart of the regime map: the modes differ, but neither dominates on quality.

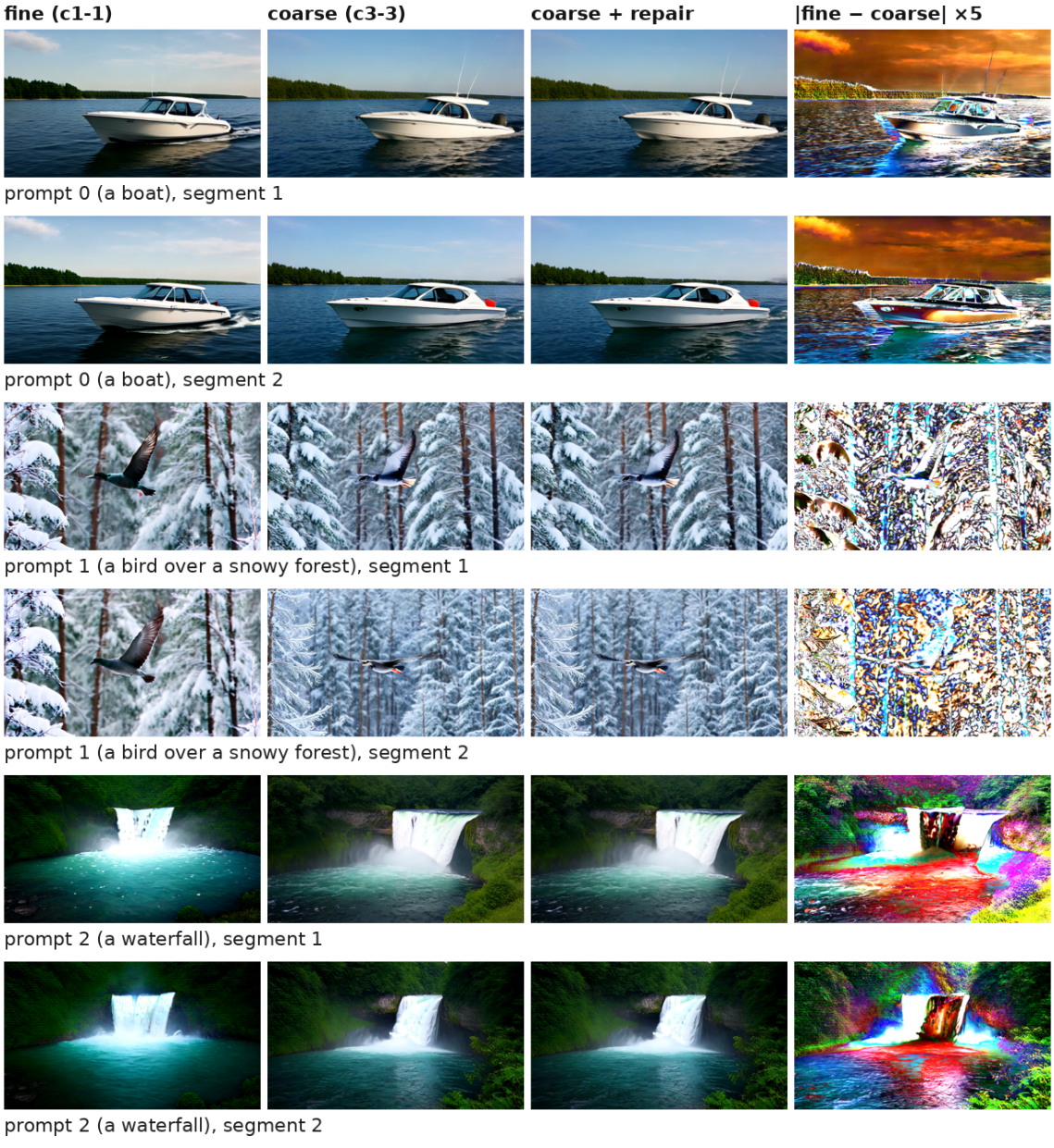


Figure 7: All generated images of the study in one frame: six matched cases (three prompts × two segments), each from the same boundary and the same noise, on T4. A segment is three latent frames (twelve pixel frames) following a shared three-frame coarse prefix. Columns: fine ($c_{1\text{-}1}$, two steps), coarse ($c_{3\text{-}3}$, two steps), coarse with the 25 % seam repair, and the absolute difference between fine and coarse amplified ×5. The modes differ in content — the bird pose and the waterfall foam both change — yet no independent endpoint ranks coarse as worse. The matched rollout videos (all policies, both GPUs) are public at https://www.kaggle.com/datasets/ajjisan/jev-adaptive-granularity-videos.

6.8. Boundaries from the remaining experiments

Several further experiments bound the scope of the method; Table 5 summarises them so that every claim in the abstract and in Fig. 1 points to a section or table.

Table 5: Remaining experiments and the boundary each establishes. All values are from the stored result files. "Status" is the disposition of the naive formulation.

experiment	result	status
decision-model critic	a Jev critic shown only pixels is at chance: discontinuity 0.497, plausible 0.514, combined 0.500 (perm $p \geq 0.41$)	open
online policy (T4, A100)	the probe-routed policy never beats the best fixed mode; the oracle router is never cost-optimal	boundary
intermediate band (T4 s_2/s_1)	coarse 27% cheaper and the latent endpoint does not penalise it, so the best policy is still fixed coarse	boundary
closed loop (T4, 48 pairs)	free post-generation evidence alone 0.725; prior + previous-segment evidence 0.690 → 0.737	bounded
escalation gate	NO-GO: the cost model, not the signal, binds	boundary

7. Boundaries and guidance

Table 6 lists the boundaries this study establishes, each with its resolution or its status. We treat a boundary as a result: it delimits the region in which a formulation is valid.

Table 6: Boundary ledger. Each entry is a measured limit of a naive formulation, with its resolution or status and the section or table that establishes it.

boundary (measured)	resolution	status	where
hand-crafted context statistics do not predict hard segments (0.491, prior report)	intrinsic prefill probe (0.798, this work)	resolved	§4.2; §6
switching costs a boundary discontinuity	training-free repair (−25%)	resolved	§4.5; Table 2
one-coarse chain stays elevated on T4 (coarse error and/or switch seam)	quantified: linear, not compounding	characterised	§6; Fig. 5
signals do not transfer across GPUs (0.798 → 0.143)	signals are regime properties	open	§6; Fig. 3
a pixel-only critic is at chance (0.497, this work)	critics need context, not pixels	open	§6.8; Table 5
the repair raises interior flicker	multi-endpoint protocol required	open	§4.5; Table 2

A practitioner recipe. (1) Measure r and Δ on the target accelerator and schedule. (2) If $r > 1$ use fine; if $r < 1$ and $\Delta \leq 0$ use coarse; only in the remaining region deploy a router. (3) If a router is deployed, prefer an intrinsic probe, calibrate its threshold on the target GPU, and never certify the result with the endpoint used to fit it.

Limitations and threats to validity. The paper's own thesis is that everything here is regime-dependent, so the scope of the claims matters. (i) *One backbone*: a single causal distilled diffusion model (Wan2.1-1.3B, Causal-rCM); the regime map is a method, not a claim about other backbones. (ii) *Two accelerators, one schedule family*: the probe inversion (0.798 → 0.143) is direct evidence that *no* number here should be assumed to transfer. (iii) *Small samples and*

a single seed: the A100 drift arm uses $n = 8$ prompts against $n = 32$ on T4, so the intervals are wide. (iv) *Asymmetric endpoint coverage*: the decoded-pixel endpoint was run on T4 only. (v) *No step-allocation or chunk-shape baseline*: the closest published alternatives are discussed but not re-implemented. (vi) *One signal family*: only prefill activations and sampler update magnitudes are probed. None of these affects the central negative result — the coarse mode was not worse in quality at any measured operating point — but each bounds its generalisation.

Generalisation. The *procedure* generalises: measure r and Δ on the target (accelerator, schedule); audit any candidate signal *per GPU* and by direct AUROC; prefer a multi-feature model when single signals invert; never certify with the endpoint used to fit. The *numbers* do not.

Ablations. From stored results: the 30-block sweep underlying Fig. 3 gives AUROC 0.71–0.80 for every block on T4, so the result is not an artefact of block 11; and the escalation-rate sweep underlying Table 3 traces the whole ρ–quality frontier, where no point beats the fixed policy.

8. Conclusion

Adaptive temporal granularity for causal video world models is not a universal win; it is a decision that depends on the accelerator, the schedule and the endpoint. We measured the three preconditions on two GPUs, built an intrinsic signal that needs no external oracle, repaired the switch seam without training, and derived a regime map that states exactly when a router is worth deploying. At every operating point we measured, the coarse mode was never worse in quality — on the latent endpoint for both GPUs and on the decoded-pixel and flicker endpoints for T4 — so a fixed policy is optimal under the stated objective and the adaptive region is empty. Where the coarse mode was also dearer (T4 s_2/s_2), fine is the cost-optimal choice even though coarse is closer to the reference. We regard the boundary analysis — the non-transfer of signals, the linearity of drift, and the context requirement of decision-model critics — as the paper's most reusable output, because each boundary is a design constraint that can be checked before deployment.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research received no external funding.

Ethics statement

Not applicable. The study involves no human participants, no animals and no personally identifiable data; all generated content is model output from public prompts.

Data availability

All code, figures and audited result files are publicly available at https://github.com/j1s4nn/JevGAG, and the consolidated result bundles are published as public datasets on Kaggle.

CRediT authorship contribution statement

Md Jisan Hossen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

Biography



Md Jisan Hossen is an undergraduate student of Artificial Intelligence at Nanjing University of Information Science and Technology (NUIST), Nanjing, China, where he started in September 2023 and is expected to graduate in June 2027. His research interest is machine vision for embodied intelligence. He is the author of the prior study on Jev-routed temporal granularity for causal video generation.

References

[1] Adnan, M., Kurella, N., Arunkumar, A., Nair, P.J., 2025. Foresight: Adaptive layer reuse for accelerated and high-quality text-to-video generation. URL: https://arxiv.org/abs/2506.00329, arXiv:2506.00329.

[2] Chen, J., Liang, Y., Liu, Z., 2026. Dflash: Block diffusion for flash speculative decoding. URL: https://arxiv.org/abs/2602.06036, arXiv:2602.06036.

[3] Geuter, J., Cheng, A., Alvarez-Melis, D., 2026. Releasing the inner jev: A simple recipe for open decision models. URL: https://arxiv.org/abs/2610.17205, arXiv:2610.17205.

[4] Graves, A., 2016. Adaptive computation time for recurrent neural networks. URL: https://arxiv.org/abs/1603.08983, arXiv:1603.08983.

[5] Hossen, M.J., 2026. Jev-routed temporal granularity with recovery and hysteresis for causal video generation. Preprint. URL: https://doi.org/10.5281/zenodo.23395414, doi:10.5281/zenodo.23395414.

[6] Huang, X., Li, Z., He, G., Zhou, M., Shechtman, E., 2025. Self forcing: Bridging the train-test gap in autoregressive video diffusion. URL: https://arxiv.org/abs/2506.08009, arXiv:2506.08009.

[7] Huang, Z., He, Y., Yu, J., Zhang, F., Si, C., Jiang, Y., Zhang, Y., Wu, T., Jin, Q., Chanpaisit, N., Wang, Y., Chen, X., Wang, L., Lin, D., Qiao, Y., Liu, Z., 2024. Vbench: Comprehensive benchmark suite for video generative models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 21807–21818.

[8] Le, T.T., Zhao, Y., Chai, M., Shen, Z., Cao, Z., Tang, D., Xie, X., Kong, D., 2026. Dsa: Dynamic step allocation for fast autoregressive video generation. URL: https://arxiv.org/abs/2606.04432, arXiv:2606.04432.

[9] Li, Y., Gao, H., Zhao, F., Tan, C., Hui, Q., Liu, S., Shi, F., Lu, T., Zhao, S., Guo, X., Su, X., Zhang, J., Dai, X., Wang, K., Lian, S., 2026. Is multimodal speculative decoding ready for diffusion-based parallel drafting? a survey and empirical diagnosis. URL: https://arxiv.org/abs/2608.20743, arXiv:2608.20743.

[10] Li, Y., Tian, C., Xia, R., Liao, N., Guo, W., Yan, J., Li, H., Dai, J., Li, H., Yang, X., 2025. Learning adaptive and temporally causal video tokenization in a 1d latent space. URL: https://arxiv.org/abs/2505.17011, arXiv:2505.17011.

[11] Li, Y., Wagle, J., 2026. Llm-as-jev: Llms are already jev-style decision models – when and how to fine-tune them. URL: https://arxiv.org/abs/2610.02076, arXiv:2610.02076.

[12] Ling, G., Xue, M., Ye, Z., 2026. Jev in the wild: A data-driven analysis of the jev model's functionality, applications and ecosystem. URL: https://arxiv.org/abs/2609.30216, arXiv:2609.30216.

[13] Liu, T., Li, Y., Lv, Q., Liu, K., Zhu, J., Hu, W., Sun, X., 2024. Pearl: Parallel speculative decoding with adaptive draft length. URL: https://arxiv.org/abs/2408.11850, arXiv:2408.11850.

[14] Ma, X., Berner, J., Liu, C., Vahdat, A., Nie, W., Wang, X., 2026a. Flex-forcing: Towards a unified autoregressive and bidirectional video diffusion model. URL: https://arxiv.org/abs/2607.03509, arXiv:2607.03509.

[15] Ma, Y., Zheng, X., Xu, J., Xu, X., Ling, F., Zheng, X., Kuang, H., Li, H., Wang, X., Xiao, X., Chao, F., Ji, R., 2026b. Flow caching for autoregressive video generation. URL: https://arxiv.org/abs/2602.10825, arXiv:2602.10825.

[16] Robitza, W., 2026. Jevqa - video quality from metadata, bitstream, and pixel features with a general-purpose decision model. URL: https://arxiv.org/abs/2609.24395, arXiv:2609.24395.

[17] Sand.ai, Teng, H., Jia, H., Sun, L., Li, L., Li, M., Tang, M., Han, S., Zhang, T., Zhang, W.Q., Luo, W., Kang, X., Sun, Y., Cao, Y., Huang, Y., Lin, Y., Fang, Y., Tao, Z., Zhang, Z., Wang, Z., Liu, Z., Shi, D., Su, G., Sun, H., Pan, H., Wang, J., Sheng, J., Cui, M., Hu, M., Yan, M., Yin, S., Zhang, S., Liu, T., Yin, X., Yang, X., Song, X., Hu, X., Zhang, Y., Li, Y., 2025. Magi-1: Autoregressive video generation at scale. URL: https://arxiv.org/abs/2505.13211, arXiv:2505.13211.

[18] Song, J., Yan, W., Huang, Y., Qiu, M., Liu, H., Zhu, X., Zhong, Y., 2026. Trac: Trajectory-aware reuse and adaptive correction for efficient autoregressive video generation. URL: https://arxiv.org/abs/2610.02779, arXiv:2610.02779.

[19] Sun, T., Zhou, Y., Liu, X., Zhang, X., Jiang, H., Cao, Z., Huang, X., Qiu, X., 2021. Early exiting with ensemble internal classifiers. URL: https://arxiv.org/abs/2105.13792, arXiv:2105.13792.

[20] Teerapittayanon, S., McDanel, B., Kung, H.T., 2017. Branchynet: Fast inference via early exiting from deep neural networks. URL: https://arxiv.org/abs/1709.01686, arXiv:1709.01686.

[21] Wang, X., Yu, F., Dou, Z.Y., Darrell, T., Gonzalez, J.E., 2017. Skipnet: Learning dynamic routing in convolutional networks. URL: https://arxiv.org/abs/1711.09485, arXiv:1711.09485.

[22] Wang, Y., Han, X., Xu, M., Perez, J.C., Douratsos, Y., He, S., Zhou, Z., Zhang, F., An, Z., Perez-Rua, J.M., Loy, C.C., Xiang, T., 2026a. In-flight kv cache with clean anchors for faster autoregressive video diffusion. URL: https://arxiv.org/abs/2609.32540, arXiv:2609.32540.

[23] Wang, Y., Wang, Y., Xue, T., Qiao, Y., Wang, Y., Chen, X., Xu, C., 2026b. Pare: Pruning and adaptive routing for efficient video generation. URL: https://arxiv.org/abs/2605.27336, arXiv:2605.27336.

[24] Xiong, T., Liew, J.H., Huang, Z., Lin, Z., Feng, J., Liu, X., 2026. Evatok: Adaptive length video tokenization for efficient visual autoregressive generation. URL: https://arxiv.org/abs/2603.12267, arXiv:2603.12267.

[25] Xu, G., Hao, J., Shen, L., Hu, H., Luo, Y., Lin, H., Shen, J., 2023. Lgvit: Dynamic early exiting for accelerating vision transformer. URL: https://arxiv.org/abs/2308.00255, arXiv:2308.00255.

[26] Xu, J., Ma, Y., Zheng, X., Wang, X., Liu, S., Yan, C., Zheng, X., Ji, R., Chao, F., Liu, S., 2026. Motion-aware caching for efficient autoregressive video generation. URL: https://arxiv.org/abs/2605.01725, arXiv:2605.01725.

[27] Yoo, S., Malla, S., Choi, C., Lu, W.D., Choi, J.H., 2026. Adept: Adaptive dynamic early-exit process for transformers. URL: https://arxiv.org/abs/2601.03700, arXiv:2601.03700.

[28] Yuan, Z., Wang, S., Xie, R., Zhang, H., Fang, T., Shang, Y., Yan, S., Dai, G., Wang, Y., 2025. Dlfr-vae: Dynamic latent frame rate vae for video generation. URL: https://arxiv.org/abs/2502.11897, arXiv:2502.11897.

[29] Zeng, Z., Hong, Y., Dai, H., Zhuang, H., Chen, C., 2023. Consistentee: A consistent and hardness-guided early exiting method for accelerating language models inference. URL: https://arxiv.org/abs/2312.11882, arXiv:2312.11882.

[30] Zhao, M., Zhu, H., Yan, B., Zhou, Z., Chen, Y., Wang, H., Sun, W., Fang, Z., Xun, Z., Li, Z., Zheng, K., He, G., Yang, X., Li, C., Bao, F., Zhu, J., 2026a. minwm: A full-stack open-source framework for real-time interactive video world models. URL: https://arxiv.org/abs/2605.30263, arXiv:2605.30263.

[31] Zhao, M., Zhu, H., Zheng, K., Zhou, Z., Yan, B., Li, X., Yang, X., Li, C., Zhu, J., 2026b. Causal forcing++: Scalable few-step autoregressive diffusion distillation for real-time interactive video generation. URL: https://arxiv.org/abs/2605.15141, arXiv:2605.15141.

[32] Zheng, D., Huang, Z., Liu, H., Zou, K., He, Y., Zhang, F., Gu, L., Zhang, Y., He, J., Zheng, W.S., Qiao, Y., Liu, Z., 2025. Vbench-2.0: Advancing video generation benchmark suite for intrinsic faithfulness. URL: https://arxiv.org/abs/2503.21755, arXiv:2503.21755.

[33] Zheng, K., He, G., Zhao, M., Zhang, J., Chen, H., Chen, J., Lin, C.H., Liu, M.Y., Zhu, J., Ma, Q., 2026. Causal-rcm: A unified teacher-forcing and self-forcing open recipe for autoregressive diffusion distillation in streaming video generation and interactive world models. URL: https://arxiv.org/abs/2606.25473, arXiv:2606.25473.

[34] Zhong, T., Tian, X., Jiang, B., Wang, X., Tao, X., Wan, P., Zhang, Z., 2025. Vfrtok: Variable frame rates video tokenizer with duration-proportional information assumption. URL: https://arxiv.org/abs/2505.12053, arXiv:2505.12053.

[35] Zhou, W., Xu, C., Ge, T., McAuley, J., Xu, K., Wei, F., 2020. Bert loses patience: Fast and robust inference with early exit. URL: https://arxiv.org/abs/2006.04152, arXiv:2006.04152.

[36] Zhu, C., Zhu, J., Li, Y., Wu, M., Song, B., Chen, C., Wu, J., Chu, X., Wang, Y., 2026a. Artifact-aware evaluation for high-quality video generation. URL: https://arxiv.org/abs/2601.20297, arXiv:2601.20297.

[37] Zhu, H., Zhao, M., He, G., Su, H., Li, C., Zhu, J., 2026b. Causal forcing: Autoregressive diffusion distillation done right for high-quality real-time interactive video generation. URL: https://arxiv.org/abs/2602.02214, arXiv:2602.02214.

[38] Zsoldos, P., 2026. Evaluating the effect of compression on video temporal consistency using objective quality metrics. URL: https://arxiv.org/abs/2605.18378, arXiv:2605.18378.